Project 1 - Explainable AI in Radiology

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Abstract

This study applies explainable AI techniques to detect pleural effusion in chest X-rays using a DenseNet169 model. Saliency Maps highlight image regions relevant to predictions, while Example-Based Explanations retrieve similar cases for comparison. Explanation quality is assessed using Quantus for Saliency Maps and Normalized Discounted Cumulative Gain (nDCG) for case rankings. The explanations presented in this work show good initial results but would still require improvement before being applied in the field.

Keywords: Chest X-Ray, Pleural Effusion, Explainable AI, Saliency maps, GradCAM, Example-based explanations, Cosine Similarity, nDCG, DenseNet

1. Introduction

Machine learning is increasingly used to solve complex problems across domains. In healthcare AI supports diagnostic and treatment decisions (9). However, many models act as black boxes with limited insight into how they generate outputs (2), making explainable AI essential to improve transparency and trust. X-ray analysis is one of the main applications of AI in medicine, especially for detecting bone and lung diseases (14). This paper presents a machine learning model to predict pleural effusion from CXR images.

2. Methods & Materials

Dataset and Preprocessing

The dataset used for this study was a smaller version of the CheXpert dataset (5), containing frontal and lateral chest X-rays, of which only the frontal images were used. Among the pathology labels available, particular attention was paid to *Pleural Effusion*, which served as the classification target.

Images with missing or uncertain *Pleural Effusion* labels were removed to improve reliability and all images were resized to 224×224 pixels and normalized to the $[0, 1]$ range. To address the class imbalance, class weights were calculated based on label frequencies and applied during training, where an 80-20 train test split was used. Since the original X-rays were grayscale, each image was duplicated across three channels to match the input format required by ImageNet-pretrained models.

Model and Training

A **DenseNet169** (12) convolutional neural network was used for the *Pleural Effusion* classification. The model was initialized with ImageNet (8) weights, freezing the lower layers while fine-tuning upper layers. A global average pooling layer was added, followed by dense layers with ReLU activations. The final layer used a sigmoid activation for binary classification. Training was performed using the Adam optimizer with a learning rate of 0.0001, binary cross-entropy loss, and class weighting. The model was trained for 10 epochs and saved.

Explanations

For this experiment two different explanations were exploited. This report explores Gradient-weighted Class Activation Mapping (**Grad-CAM**), which identifies critical regions in an image by leveraging gradients in the last convolutional layer of a CNN and global-average-pooling to assess neuron importance (15). Grad-CAM generates heatmaps that highlight relevant features for model predictions. To evaluate the quality of saliency maps, the Quantus toolkit (11) was used, employing metrics such as Pixel-Flipping

(6) (Faithfulness), Local Lipschitz Estimate (3) (Robustness), Model Parameter Randomisation (1) (Randomisation), Relevance Mass Accuracy (4) (Localization), and Complexity (7) (Complexity).

For example-based explanations, image feature embeddings from the last pooling layer are extracted and ranked using **cosine similarity**, measuring vector orientation and magnitude (10). These rankings are compared to expert-defined rankings via Normalized Discounted Cumulative Gain (**nDCG**) (13), which assesses ranking quality by assigning relevance scores in the range $[5.5, 1]$, computing discounted gain, and normalizing against the ideal expert ranking. This ensures that the model groups similar images in a clinically meaningful way.

3. Results & Discussion

The following sections present and discuss the results of testing the model and explanation methods from Section 2 on predefined test images.

Model Training and Evaluation

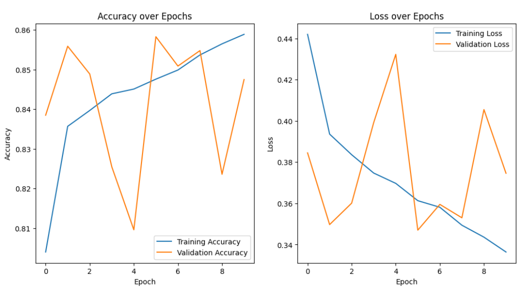

Figure 1: Training and validation per epoch

During training, the DenseNet169 model steadily improved in training accuracy and loss, though validation accuracy fluctuated, indicating some overfitting (Figure 1). The final model achieved 79.21% accuracy on the test set, with a precision of 63.41% and recall of 81.25%, showing good detection of positive *Pleural Effusion* cases. The F1-score was 71.23%, balancing precision and recall, while the ROC AUC reached 84.96%, demonstrating strong overall discriminatory power. A binomial test yielded a p-value of 2.03×10^{-17}, confirming the performance is significantly better than chance.

Saliency Maps

Figure 3 shows the GradCAM explanations generated to gain insights about the models' focus on the test images. The most important region, as identified by the model, is represented in seismic colors: red, white, and blue. The bounding boxes on the explanation boxes circle the critical parts of the image, that the explanation should focus on. In all five test images, the explanations highlight parts of the bounding

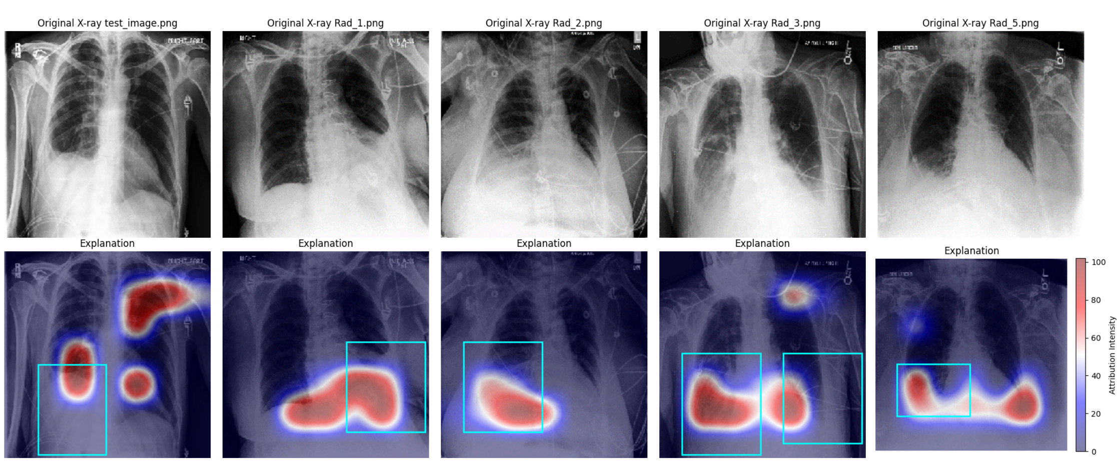

Figure 2: GradCAM Explanations

box but also often highlight irrelevant areas outside of it.

In most cases, the heatmaps focus on the lower lung regions associated with Pleural Effusion, suggesting the model identifies clinically relevant areas. However, some explanations highlight unrelated regions, indicating that the model occasionally relies on uninformative features to classify Pleural Effusion.

Additionally, Table 3 shows the mean and standard deviation of the Quantus evaluation metrics on the test images [1].

	Mean	**Standard deviation**
Complexity	9.36	0.24
Local Lipschitz	0.03	0.03
Pixel Flipping	9.93	0.49
Localisation	0.25	0.08

Table 1: Quantus: Explanation Evaluation

The complexity score (9.36) confirms the high complexity of CNN explanations, nearing the upper bound based on feature usage. The robustness score (0.03) shows the model is highly stable to small perturbations. The pixel flipping score (9.93) indicates that removing important pixels significantly reduces confidence, reinforcing the explanation's reliability. The localization score (0.25) suggests a moderate alignment between highlighted regions and the model's decision-making process, with some imprecision in feature attribution.

Example-based Explanations

The results show strong alignment between the scientists' and cosine similarity rankings. The high nDCG score (0.97) confirms that extracted features correlate well with expert judgment. While the line plot (Figure 3) reveals some ranking mismatches, overall similarity remains consistent, especially in the top-ranked images. The heatmap in plot (Figure 4) further shows that five of the top six images appear in both rankings, though in a slightly different order.

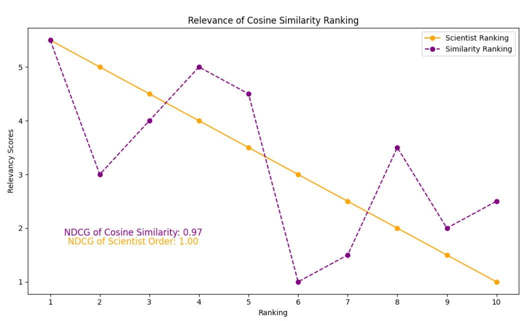

Figure 3: The line plot of images ranked by the scientists and cosine similarity.

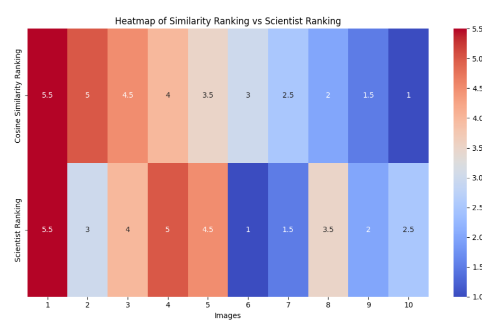

Figure 4: heatmap of the similarity rankings (identical images share the same color)

4. Conclusion

Understanding the decision process of machine learning models is key to transparency and trust, making explainability essential in medical AI systems. This study assessed two explanation methods: Grad-CAM, which highlights important regions in X-rays but can suffer from diffuse attributions, and example-based explanations, which retrieve similar cases via cosine similarity but require a well-curated dataset. Grad-CAM is more effective for clinicians seeking localized insights, while similarity-based explanations aid comparative analysis and proved to be more satisfying in terms of precision. Further improvements and model comparisons should be explored, preferably using PyTorch for better integration with Quantus.

1. Randomisation was excluded due to practical issues—we were unable to make it stop running and produce relevant results. We suspect this might be due to an incompatibility the model and Quantus's randomization process.

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